

# SINAIS

A Proposed Framework to Assess Reliability, Flexibility & Resilience of Power Systems

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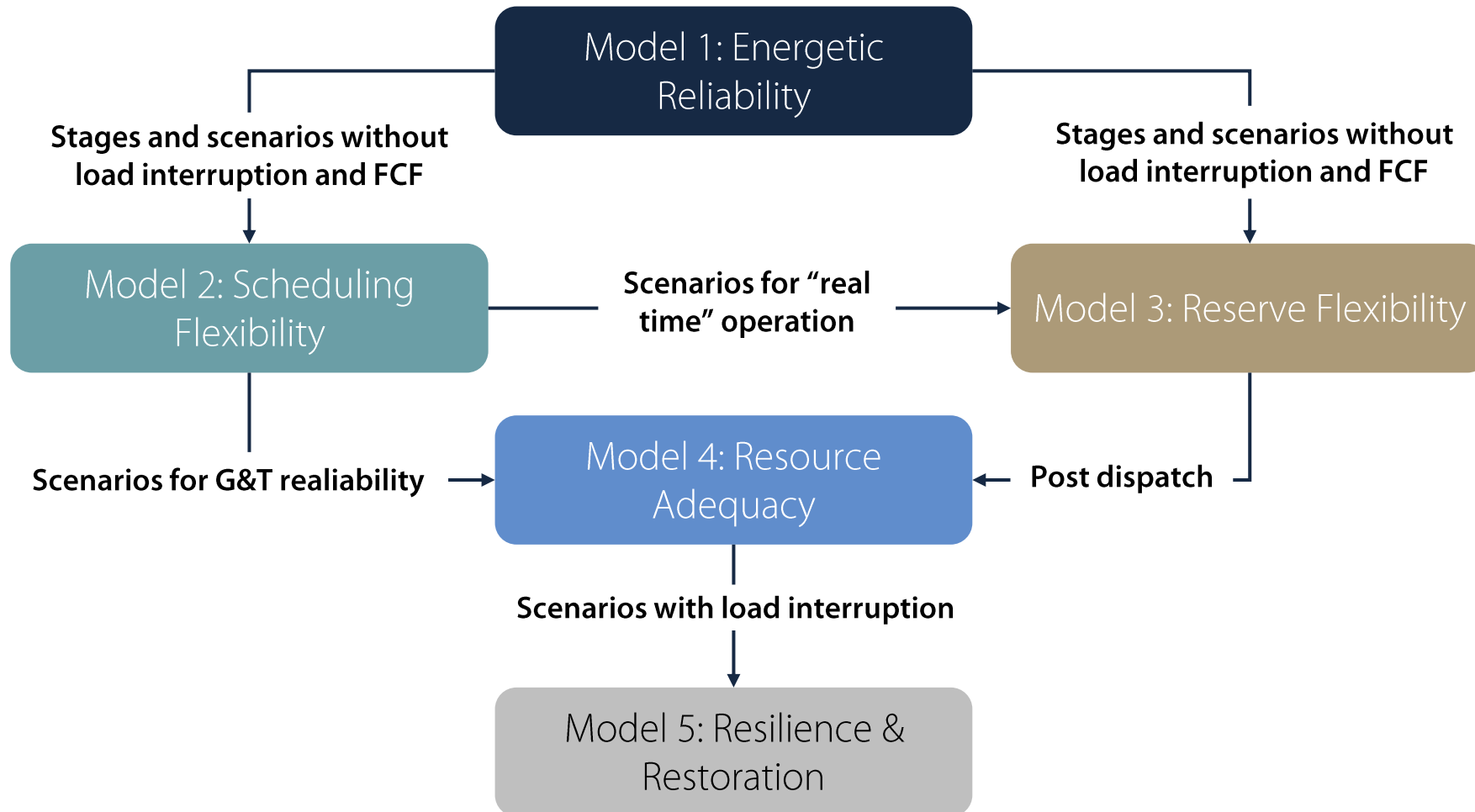
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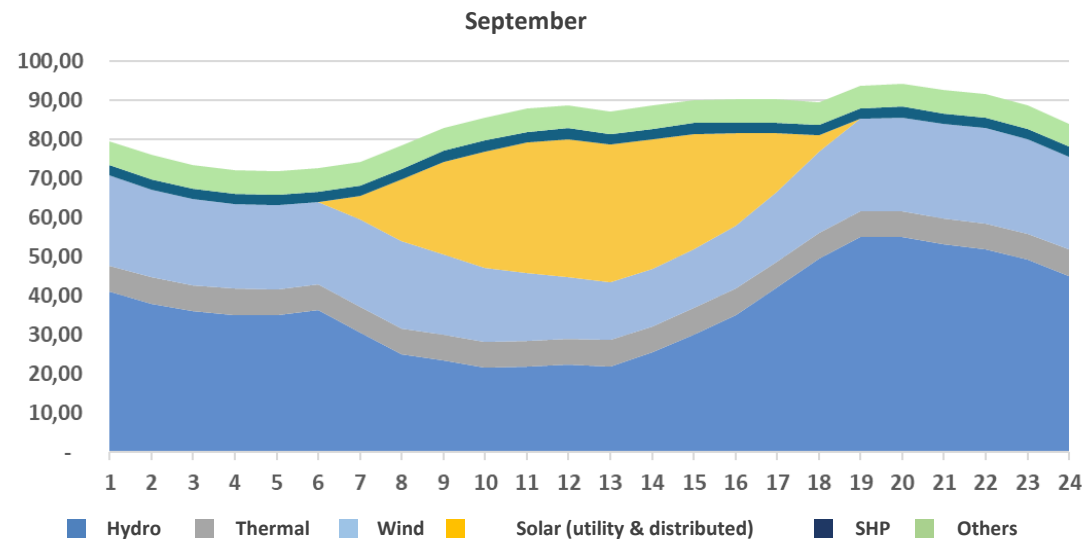
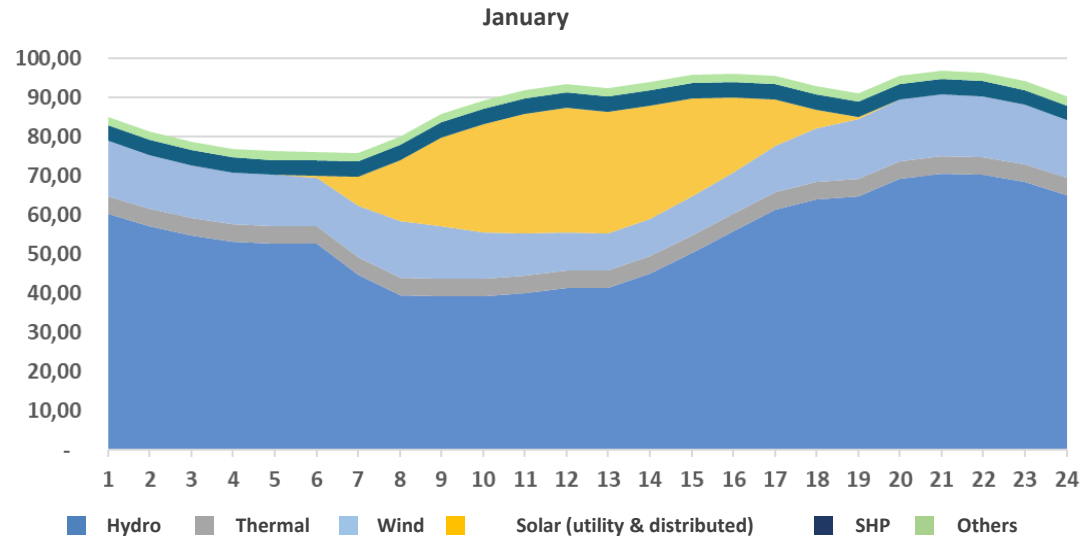
# SINAIS – SYSTEM OF INDICATORS FOR ANALYSIS OF INTERCONNECTED SYSTEMS

## RELIABILITY, FLEXIBILITY AND RESILIENCE



# SINAIS – Energetic Reliability

## Model 1: Energetic Reliability



Historical data

Irradiation

Temperature

Wind

Inflows

Water Inflows & Renewable Scenarios



Operational planning



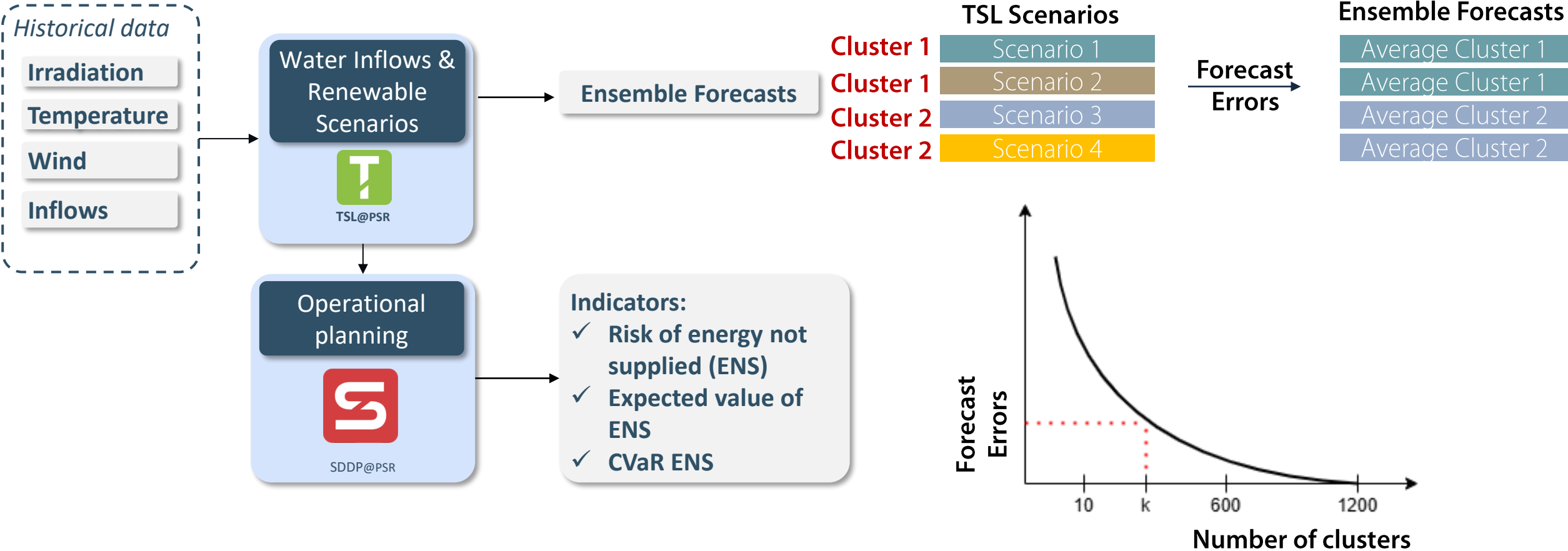
SDDP@PSR

Indicators:

- ✓ Risk of energy not supplied (ENS)
- ✓ Expected value of ENS
- ✓ CVaR ENS

# SINAIS – Scheduling Flexibility

Model 2: Scheduling Flexibility



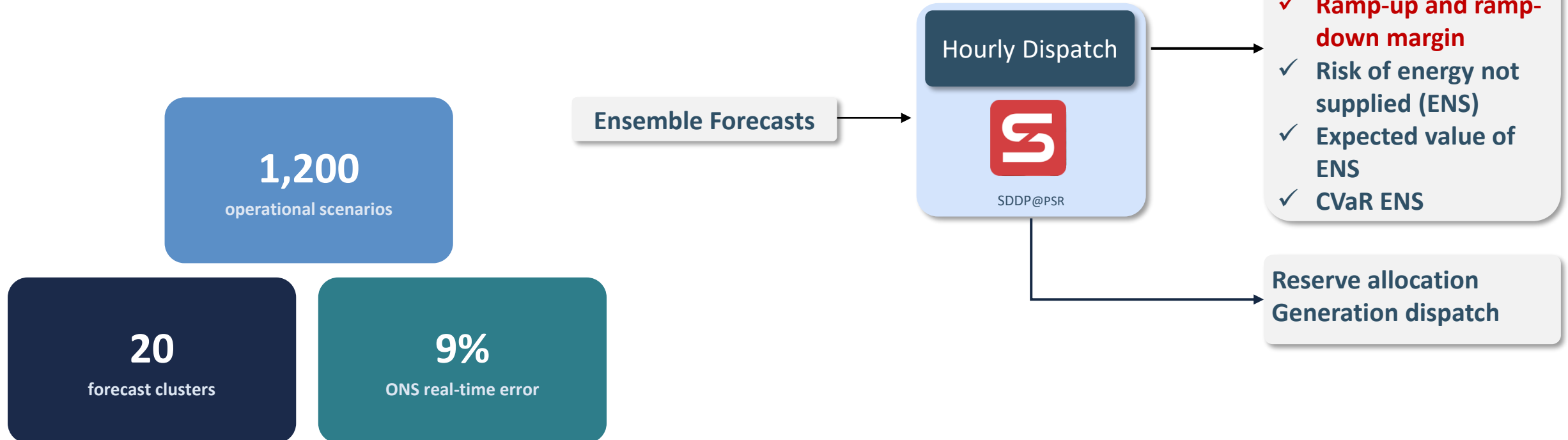


# SINAIS – Scheduling Flexibility

Model 2: Scheduling Flexibility

**K-means clustering** builds "ensemble forecasts" whose **spread** matches the **operator's real-time forecast error**, so flexibility is neither over- nor under-stated

Hourly SDDP dispatch then produces generation and reserve schedules for every scenario — the raw material for all indicators



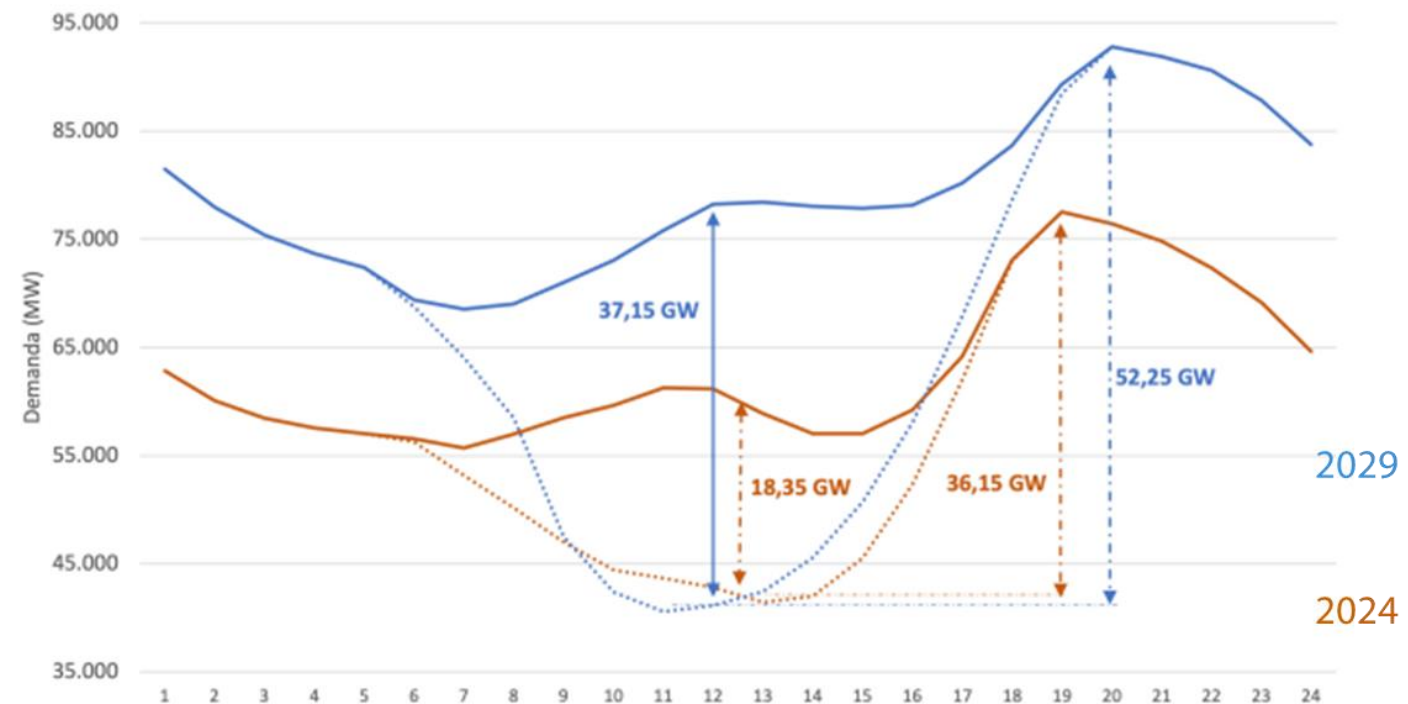
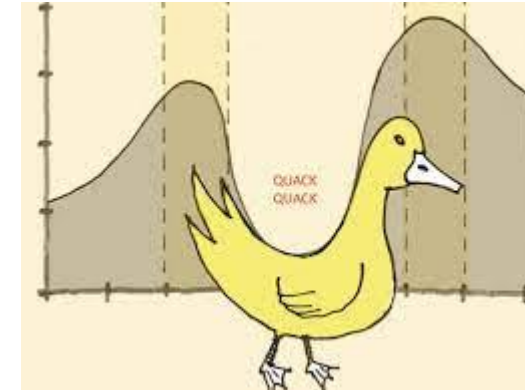
# Scheduling flexibility: can we follow the net-load ramp?

**Controllable plants — mainly hydros — must serve the net-load curve.**

As solar rises at midday, net demand falls sharply (down-ramp); as it sets, net demand climbs into the evening peak (up-ramp).

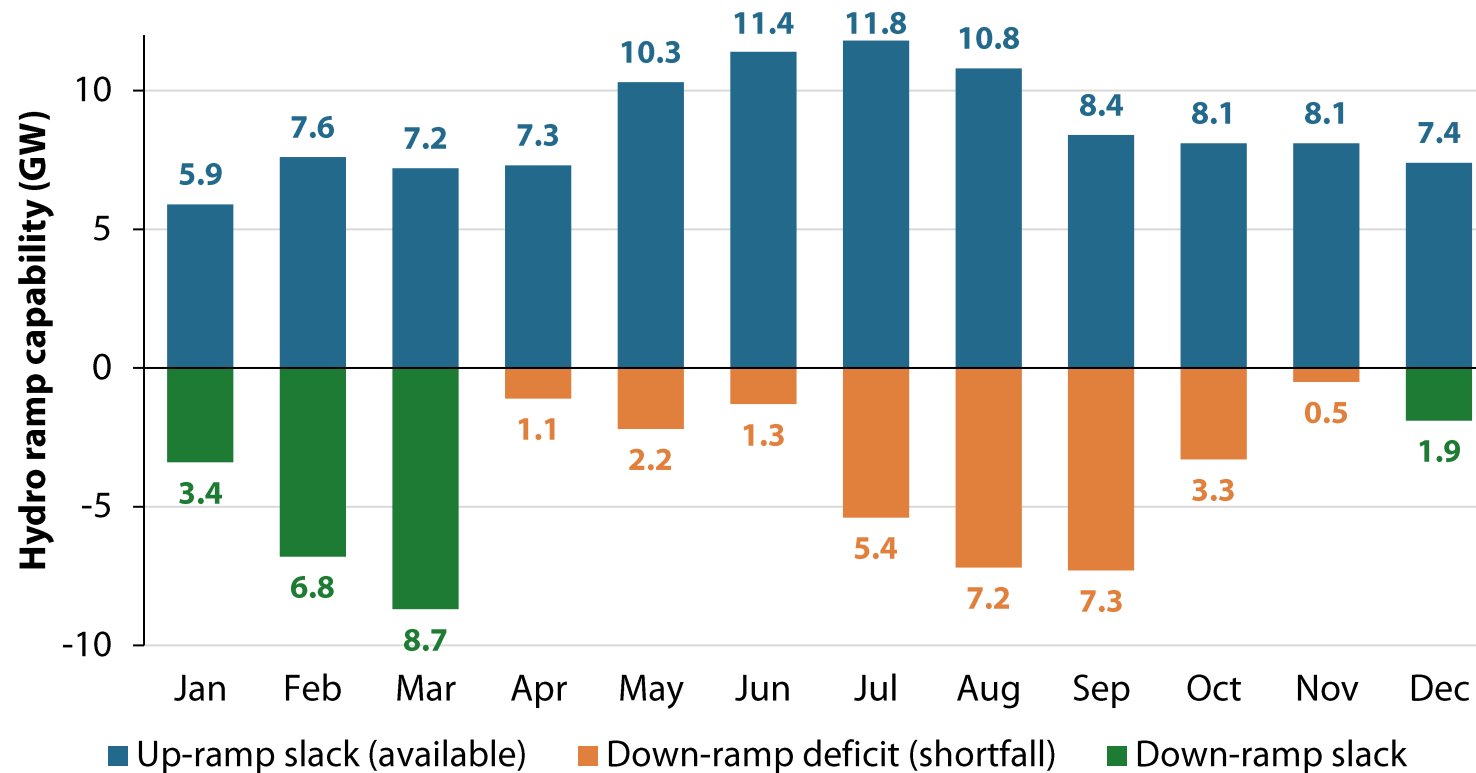
For every hour we compare available hydro ramp against the required ramp, yielding four indicators:

- Up-ramp slack • Up-ramp deficit
- Down-ramp slack • Down-ramp deficit



Illustrative net-load shape (Brazil, spring). Slack = spare ramp; deficit = ramp shortfall

# Hydro ramp capability vs. net-load requirement



Up-ramp slack is positive all year (5.9–11.8 GW): the fleet can always ramp up

**A down-ramp deficit appears from April and peaks in Jul–Sep (up to 7.3 GW) – forcing renewable curtailment**

This coincides with negative marginal costs in the Northeast: spilled renewable energy

14.15 TWh of renewable energy curtailed in 2026

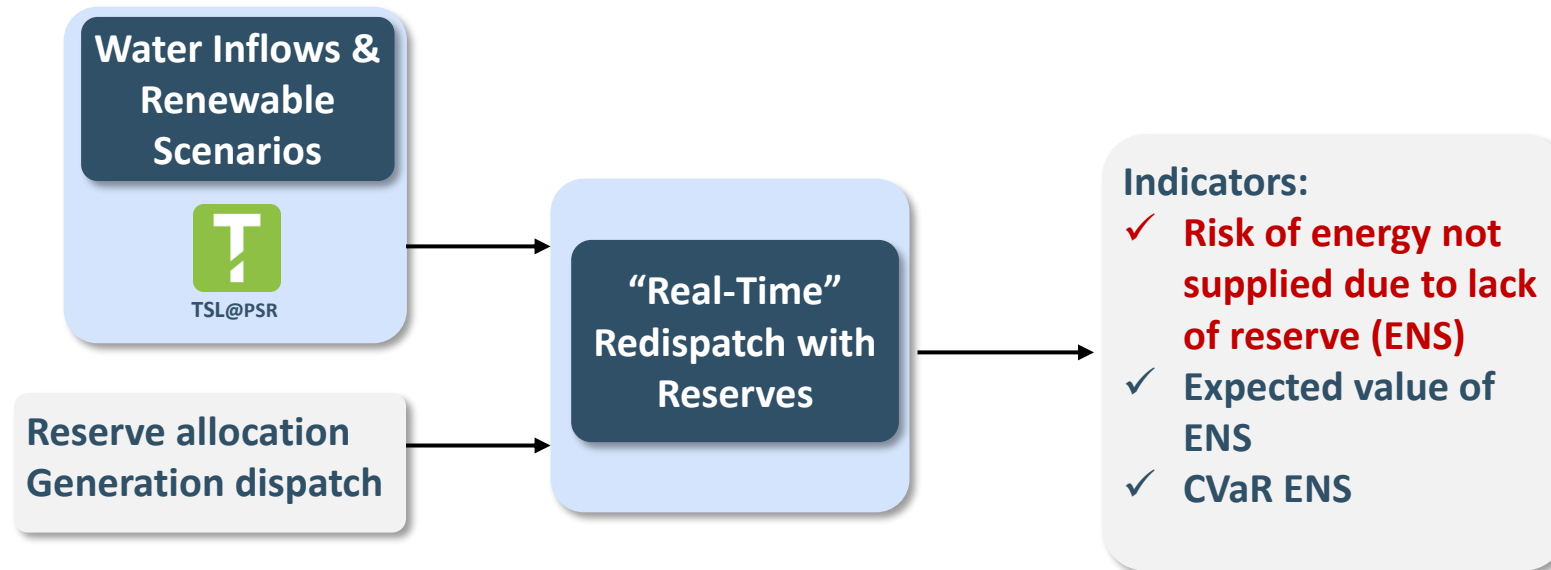
Estimated economic value: USD 414 million

**7.3 GW**

peak downward-ramp deficit (September)

# SINAIS – Reserve Flexibility

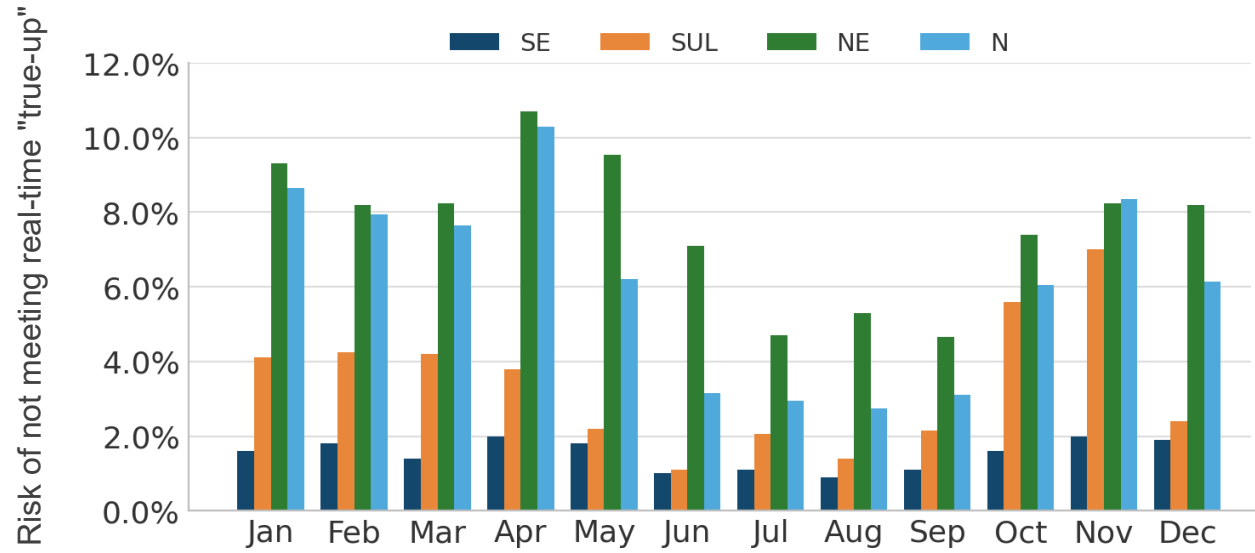
## Model 3: Reserve Flexibility





# Reserve flexibility: real-time “true-up” sufficiency

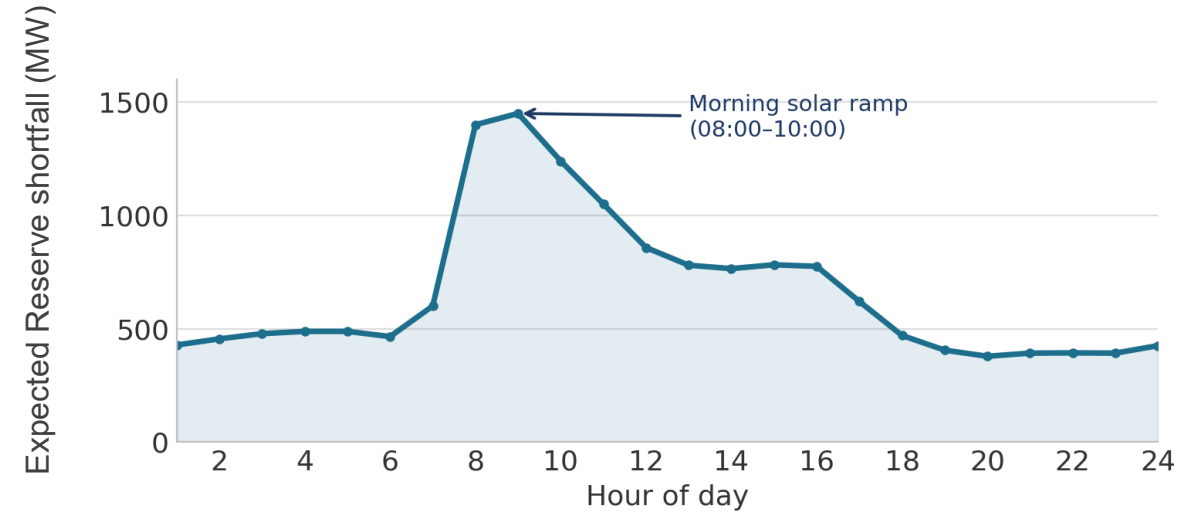
## Can scheduled reserves absorb forecast errors?



This result highlights the difficulty these subsystems face in absorbing deviations between forecast and actual renewable generation using the available reserves

Northeast and North carry the highest risk of not accommodating real-time deviations — up to ~10% in April.

These subsystems have high variable-renewable share and few units on AGC



*Intraday, the reserve gap peaks at 08:00–10:00 as solar ramps up and displaces flexible units.*

**~300 MW**

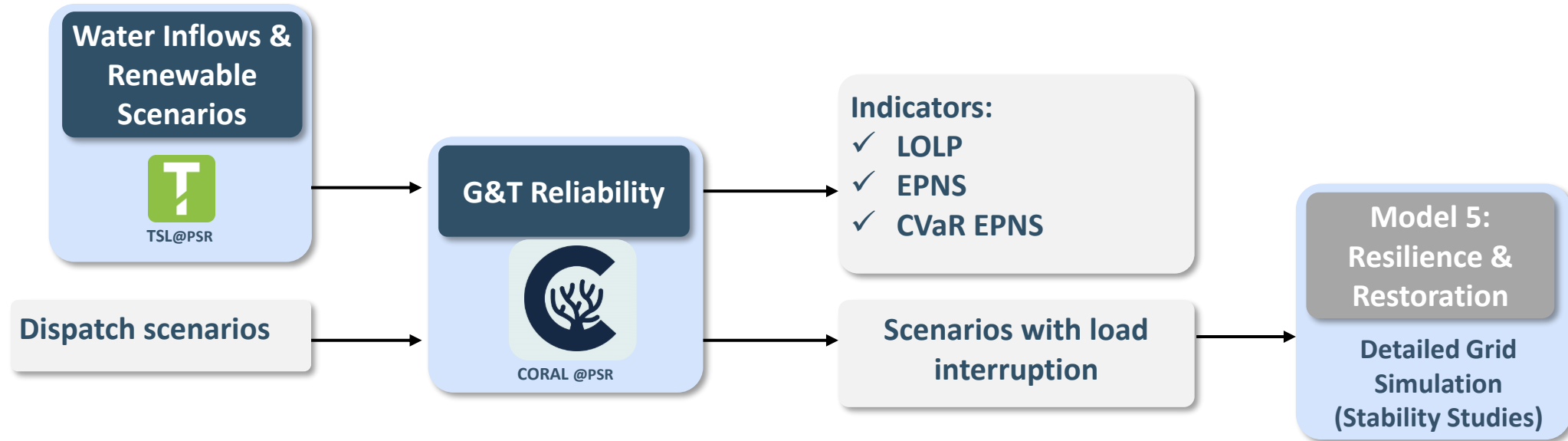
average reserve shortfall  
(North, April)

**~1,450 MW**

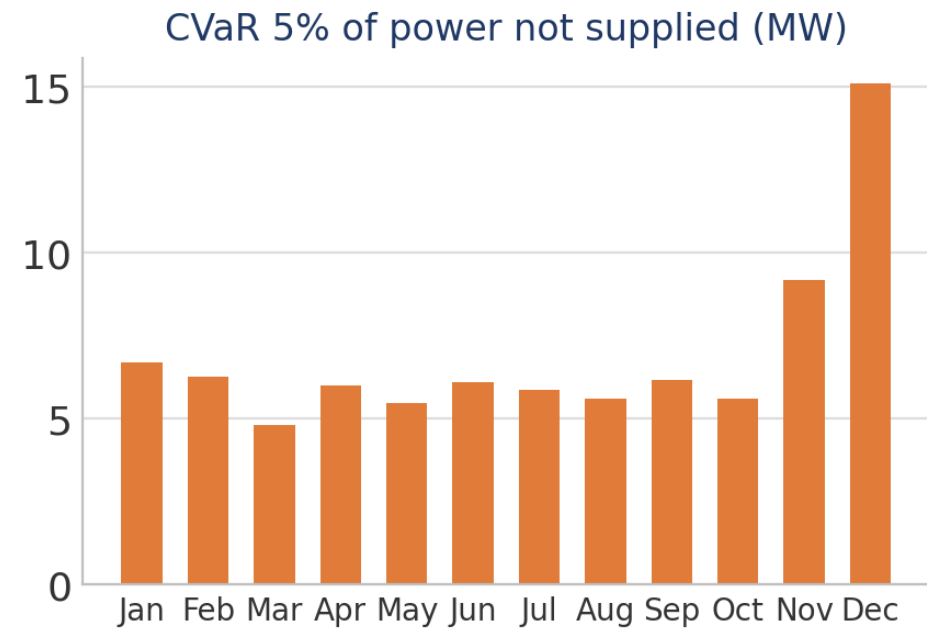
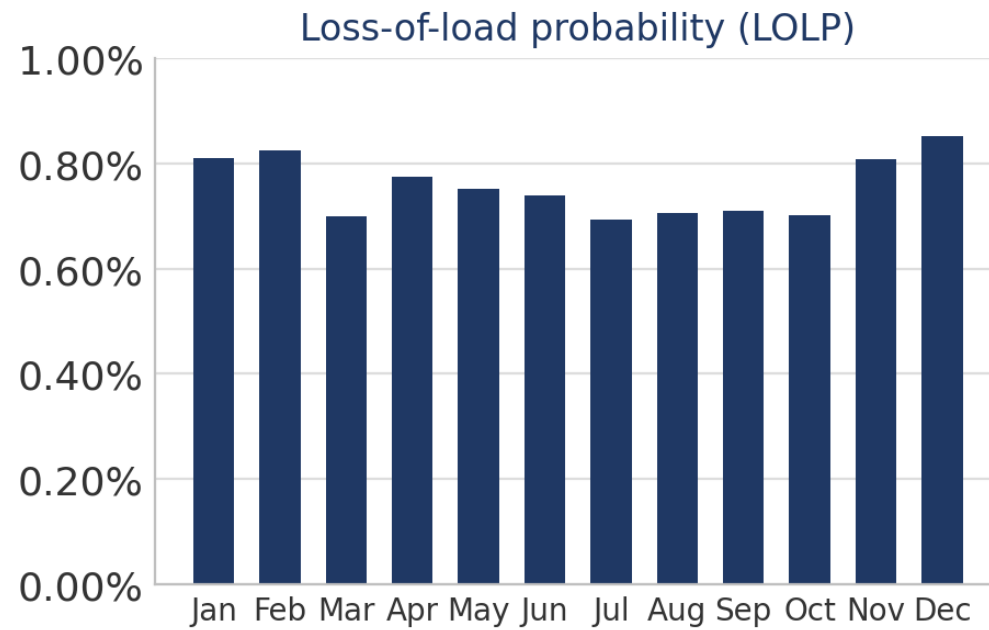
morning reserve gap  
(April, 08–10h)

# SINAIS – Resource Adequacy

## Model 4: Resource Adequacy



# SINAIS – Resource Adequacy



# Key Takeaways



**Energy reliability needs a truly stochastic PCM:** with the strong renewable penetration now seen in the US, the production-cost model must compute the system operating policy and storage opportunity costs accurately



**One framework for reliability, flexibility and resilience:** five linked procedures provide a multidimensional, forward-looking view of system health



**One integrated platform — fully automatable:** the entire framework runs end-to-end on a single platform, so it can be automated into live dashboards that support decision-making



**Flexibility is often the binding constraint:** under high renewables the limits show up as downward ramp and real-time reserve sufficiency — not necessarily energy or peak capacity



# THANK YOU!

## Any questions?

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